La Scuola d'Italia / Dr. Huson / IB Math 11
26 October 2026

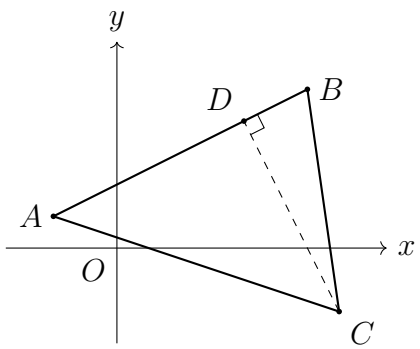
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2.4 Pre-Quiz: Lines, Exponents and Radicals

A calculator is allowed. When a question asks for an exact answer or a particular form, show the algebra that leads to it.

1. The points $A(-2,\ 1)$, $B(6,\ 5)$ and $C(7,\ -2)$ are the vertices of triangle ABC. The line L passes through C and is perpendicular to the line AB. The line L meets the line AB at the point D.

   (a) Find the gradient of the line AB.
   (b) Find the equation of L. Give your answer in the form $y = mx + c$.
   (c) Find the coordinates of D.
   (d) Find the length CD. Give your answer in the form $a\sqrt{5}$, where $a \in \mathbb{Z}$.
   (e) Find the area of triangle ABC.



2. Company A hires out cars. It charges a fixed fee plus an amount for each day. Hiring a car for 3 days costs 165 euros, and hiring it for 7 days costs 305 euros. The cost, in euros, of hiring a car for d days from company A is $C_A(d) = md + c$.

   (a) Find the value of m.

   (b) Find the value of c.

   (c) Interpret the value of c in the context of the question.

   Company B charges 10 euros plus 45 euros for each day, so $C_B(d) = 45d + 10$.

   (d) Find the number of days for which the two companies charge the same amount.

   (e) Mia wants to hire a car for 9 days. State which company is cheaper, and find how much she saves by choosing it.

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3. (a) Solve the system of equations

$$3x - 4y = 18, \qquad 5x + 2y = 4.$$

The line L_1 has equation $2x+ky = 6$, where $k \neq 0$. The line L_2 has equation $3x-6y = 5$.

(b) Find the value of k for which L_1 is parallel to L_2.

(c) For this value of k, write down the number of solutions of the system $2x + ky = 6$, $3x - 6y = 5$. Give a reason for your answer.

(d) Find the value of k for which L_1 is perpendicular to L_2.

(e) For the value of k in part (d), find the exact coordinates of the point where L_1 and L_2 intersect.

4. (a) Simplify $\dfrac{(2x^3y^{-2})^3}{4x^5y^{-4}}$. Give your answer using positive exponents only.

   (b) Write $(27x^6)^{-\frac{2}{3}}$ in the form ax^n, where $a \in \mathbb{Q}$ and $n \in \mathbb{Z}$.

   (c) Solve the equation $9^{x+1} = \left(\dfrac{1}{27}\right)^x$.

   Consider the equation $2(4^x) - 17(2^x) + 8 = 0$.

   (d) Let $u = 2^x$. Show that the equation can be written as $2u^2 - 17u + 8 = 0$.

   (e) Hence, solve the equation $2(4^x) - 17(2^x) + 8 = 0$.

   (f) Use your calculator to solve the equation $5^x = 40$.

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5. In this question, give every answer in exact form.

   (a) Simplify $\sqrt{75} + \sqrt{48} - 2\sqrt{27}$.

   (b) Expand and simplify $\left(2\sqrt{5} - 3\right)^2$.

   (c) Write $\dfrac{\sqrt{3}+1}{\sqrt{3}-1}$ in the form $a + b\sqrt{3}$, where $a, b \in \mathbb{Z}$.

   (d) Solve the equation $x\sqrt{2} + 3 = x\sqrt{8} - 5$. Give your answer in the form $a\sqrt{2}$.

   (e) A rectangle has length $\left(3 + \sqrt{2}\right)$ cm and area $\left(7 + 7\sqrt{2}\right)$ cm^2. Find the width of the rectangle in the form $a + b\sqrt{2}$, where $a, b \in \mathbb{Z}$.

6. The line L passes through the points $A\left(\sqrt{2},\ 3\right)$ and $B\left(3\sqrt{2},\ 7\right)$.

   (a) Show that the gradient of L is $\sqrt{2}$.

   (b) Find the equation of L in the form $y = mx + c$.

   (c) Find the exact x-intercept of L. Give your answer with a rational denominator.

   The line L meets the line $y = 2x$ at the point P.

   (d) Find the exact coordinates of P. Give each coordinate in the form $a + b\sqrt{2}$, where $a, b \in \mathbb{Q}$.

   (e) Find the exact length AB. Give your answer in simplest surd form.